

Manufacturing Quality & Yield Analysis — Synthetic Industrial Dataset Design

Project Purpose

This dataset was intentionally designed to simulate a realistic manufacturing production environment where operational performance depends on machine condition, maintenance behavior, environmental variation, process stability, and human operational factors.

The objective was not to generate random numbers, but to create a multi-table industrial system with believable operational relationships, realistic failure patterns, and imperfect data quality similar to what analysts encounter in real manufacturing environments.

The dataset supports:

- operational analytics
 - manufacturing KPI analysis
 - statistical quality studies
 - anomaly detection
 - root-cause investigation
 - dashboard reporting
 - machine learning workflows
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Design Philosophy

Most public manufacturing datasets are either:

- too clean
- too small
- disconnected from business logic
- missing operational dependencies
- unrealistic in behavior

To make the project more realistic, this dataset was built around several industrial principles:

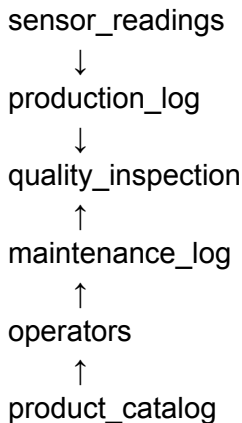
- machines degrade over time
- maintenance influences reliability
- environmental instability affects quality
- sensor drift can occur before failure
- operators influence process consistency
- inspection labels contain some uncertainty
- production systems contain messy and incomplete data

The dataset was intentionally designed to behave imperfectly under analysis so that the analytics process would resemble real operational work.

Database Architecture

The dataset follows a multi-table manufacturing schema.

Each table represents a different operational layer inside a production environment.



The production table acts as the central operational layer connecting machine activity, quality inspection outcomes, maintenance history, operator behavior, and product specifications.

1. sensor_readings Table

Purpose

The `sensor_readings` table simulates high-frequency industrial machine telemetry generated during production.

This table represents process variables continuously monitored in manufacturing systems.

Business Logic

Sensor behavior was designed to simulate realistic process instability rather than purely random measurements.

Several operational concepts were intentionally introduced:

- sensor drift before defects
- gradual degradation patterns
- unstable machine behavior
- noisy measurements
- environmental sensitivity

- missing sensor channels

The objective was to create time-series behavior that analysts could meaningfully investigate using rolling statistics, anomaly detection, and feature engineering.

Key Relationships

- Linked to machines through `machine_id`
 - Linked to production windows through timestamps
 - Used to engineer pre-defect operational signals
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Simulated Data Quality Issues

The table intentionally includes:

- ~30% null rates in selected sensor channels
- outlier spikes
- timestamp irregularities
- noisy measurements
- drift patterns before failures
- duplicated rows caused by simulated clock skew

These imperfections were added to reflect real industrial telemetry systems where sensor reliability is rarely perfect.

Analytical Purpose

This table supports:

- time-series analysis
 - anomaly detection
 - rolling-window statistics
 - pre-failure pattern analysis
 - process stability monitoring
 - feature extraction for machine learning
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2. production_log Table

Purpose

The `production_log` table represents production execution records for each manufacturing batch or production run.

This table acts as the operational backbone of the dataset.

Business Logic

Each production run was designed to simulate realistic operational conditions influenced by:

- machine assignment
- operator assignment
- shift timing
- production speed
- environmental conditions
- product specifications

Production duration and operational efficiency were influenced by multiple interacting variables rather than isolated random values.

Key Relationships

This table connects:

- machine activity
- operators
- production timing
- environmental conditions
- quality inspection outcomes

It acts as the central join layer across the entire database structure.

Operational Assumptions

Several realistic assumptions were intentionally introduced:

- certain shifts operate less efficiently
 - environmental conditions fluctuate over time
 - machine load affects operational stability
 - batch size influences production duration
 - recommended machine speed impacts defect probability
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Analytical Purpose

This table supports:

- production KPI analysis
 - operational efficiency studies
 - shift-level performance analysis
 - throughput monitoring
 - environmental risk analysis
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3. quality_inspection Table

Purpose

The `quality_inspection` table represents final inspection outcomes after production completion.

This table contains the primary target variable for quality analysis and machine learning.

Business Logic

Defects were not randomly assigned.

Failure probability was generated using stacked operational risk factors such as:

- sensor instability
- maintenance delay
- operator experience
- environmental stress
- machine degradation
- excessive production speed

The probability of failure increased when multiple operational risks occurred together.

Target Variable Design

The primary target variable:

PIN_HOLES = 0 or 1

was intentionally designed with class imbalance to simulate real industrial quality systems where failures are relatively uncommon.

Additional Quality Signals

The table also includes:

- defect counts
 - severity scoring
 - defect-type categorization
 - inspection timing
 - inspector identifiers
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Label Noise

To simulate realistic inspection inconsistency, a small amount of label noise was intentionally introduced.

This reflects real production environments where inspections are not perfectly consistent.

Analytical Purpose

This table supports:

- classification modeling
 - defect analysis
 - root-cause investigation
 - severity trend analysis
 - quality KPI reporting
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4. maintenance_log Table

Purpose

The `maintenance_log` table simulates preventive and corrective maintenance events for industrial equipment.

Business Logic

Machine reliability was intentionally linked to maintenance timing.

The simulation assumes:

- delayed maintenance increases failure risk
- preventive maintenance improves stability
- corrective maintenance occurs after operational issues
- downtime affects production efficiency

The relationship between maintenance history and production quality was designed to create measurable operational patterns during analysis.

Maintenance Types

Two maintenance strategies were simulated:

- preventive maintenance
- corrective maintenance

These categories allow comparison between proactive and reactive maintenance behavior.

Operational Effects

Maintenance timing influences:

- machine reliability
 - sensor stability
 - defect probability
 - downtime behavior
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Analytical Purpose

This table supports:

- maintenance-effectiveness analysis
 - MTBF analysis
 - MTTR analysis
 - downtime studies
 - reliability analytics
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5. operators Table

Purpose

The **operators** table represents workforce-related operational information.

Business Logic

Operator behavior was intentionally connected to production quality.

Several workforce assumptions were simulated:

- experienced operators create fewer defects
- certified operators operate more consistently
- training improves process reliability
- inexperienced operators contribute more operational variability

The table was designed to support human-factor analysis instead of treating operators as static IDs.

Privacy Design

Operator names were intentionally anonymized using hashed identifiers to simulate privacy-safe industrial reporting systems.

Analytical Purpose

This table supports:

- human-factor analysis
 - operator performance studies
 - certification impact analysis
 - experience-based segmentation
 - workforce quality monitoring
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6. product_catalog Table

Purpose

The `product_catalog` table stores contextual production specifications for each product type.

Business Logic

Different products were intentionally designed with different manufacturing tolerances.

This introduces realistic operational complexity because some products are naturally more defect-sensitive than others.

The simulation assumes:

- tighter tolerances increase quality sensitivity
 - high-speed production can exceed safe operational limits
 - different product families behave differently during production
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Operational Impact

Product specifications influence:

- defect probability
 - process sensitivity
 - machine stress
 - recommended operating speed
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Analytical Purpose

This table supports:

- product-level defect analysis
 - tolerance sensitivity analysis
 - production optimization studies
 - product-family segmentation
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Simulated Data Quality Challenges

The dataset intentionally contains realistic industrial data issues.

Included Challenges

- missing values
- duplicate rows
- timestamp drift
- outlier sensor behavior
- noisy labels
- class imbalance
- nested JSON fields
- inconsistent operational patterns

These issues were included intentionally because real industrial systems rarely produce perfectly structured data.

Why This Matters

The goal of this dataset is not simply visualization.

The project was designed to simulate how analysts work with:

- messy operational systems

- imperfect telemetry data
- production quality problems
- maintenance-driven failures
- business-oriented KPI reporting

The dataset behaves realistically under:

- EDA workflows
- dashboard analysis
- statistical testing
- machine learning
- operational investigations

This allows the project to function as a reusable industrial analytics framework rather than a simplified academic dataset.